

MAHESH YADDANAPUDI

ML Engineering Leader · Agentic AI Platform Architecture, MCP & Retrieval

Architecting agent platforms on 17+ years of enterprise engineering

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PROFESSIONAL SUMMARY

Machine learning engineering leader with 17+ years of production engineering experience, focused on architecting agentic AI platforms: multi-agent orchestration, MCP tool gateways and registries, hybrid retrieval (vector, keyword, graph), human-in-the-loop workflows, and agent evaluation/observability. Has designed and built agent platforms end to end — in Python (LangGraph, FastAPI) and Java (Spring AI, Spring Boot) — both inside a large enterprise and in open source, on a foundation of microservices, Kubernetes, and cloud architecture built at Google, Oracle, JPMorgan Chase, and Workday.

CORE COMPETENCIES

AGENTIC AI ARCHITECTURE

Multi-Agent Orchestration (LangGraph) | Model Context Protocol (MCP) | Tool & Agent Registries
Spring AI | A2A | CodeAct | Human-in-the-Loop Workflows | Docker-Sandboxed Tool Execution

RETRIEVAL & EVALUATION

Hybrid Retrieval (FAISS + BM25 + Neo4j GraphRAG) | Reciprocal Rank Fusion | Embeddings
Apache Lucene | OpenTelemetry | LangSmith | LLM-as-Judge | Tool-Call Auditing

ML & LANGUAGES

Forecasting (N-BEATS, Prophet) | ML Pipelines | MLX Fine-Tuning (LoRA/QLoRA)
Local Voice AI (Whisper, sherpa-onnx, Kokoro) | Prompt Engineering | Python | Java | TypeScript
FastAPI | Spring Boot | React | Angular

CLOUD & DEVOPS

Google Cloud Platform | AWS | Oracle Cloud | Docker | Kubernetes | Jenkins | Ansible | CI/CD
Infrastructure as Code

ARCHITECTURE & DATA

Microservices | Netflix OSS | REST APIs | SOA | PostgreSQL | Neo4j | Elasticsearch | Oracle
MySQL | SQL Server | IBM DB2

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer

Sep 2022 – Present

WORKDAY

Santa Clara, California, United States

Software Engineer, Sep 2022 – Feb 2026 · promoted to Senior ML Engineer, Feb 2026

Engineer on an enterprise planning platform, focused on ML-driven forecasting, workflow automation, and agentic AI systems.

- ▶ Architected and lead-developed an end-to-end predictive forecasting (ML) application: forecast definitions using historical actuals as reference and a target planned version as output, selectable forecasting models (N-BEATS, Prophet), and dispatch to an ML pipeline that computes forecasts and writes results back to the planned version.
- ▶ Built the forecasting scheduler as phase two: create/update and pause/resume schedules, timezone handling, OpenAPI/Swagger documentation, and expanded automated test coverage.
- ▶ Designed and led a lightweight, decoupled process/workflow engine powering configurable enterprise workflows — defining its core interfaces, lifecycle operations, and extensibility model.
- ▶ Pioneered AI-agent orchestration with the Model Context Protocol (MCP) and Spring AI — server/host integrations exposing enterprise workflow APIs as tools for AI assistants; the host design was adopted as an architectural baseline by the owning team.
- ▶ Prototyped natural-language metadata discovery with a hybrid GraphRAG approach — FAISS vector, keyword, and Neo4j graph retrieval fused via reciprocal rank fusion, with LLM intent parsing and grounded synthesis.
- ▶ Champion of AI-assisted engineering — applying LLM tooling (Claude, GPT-4, Cursor) to design, prototyping, code/test generation, and debugging, and evangelizing AI-first workflows across the team.

Principal Member of Technical Staff

ORACLE

Jan 2022 – Sep 2022

Seattle, Washington, United States

Contributed to Oracle's cloud infrastructure and enterprise solutions as a Principal Member of Technical Staff.

Software Engineer

GOOGLE

May 2021 – Dec 2021

Warsaw, Mazowieckie, Poland

Managed customer problems through effective diagnosis and resolution. Worked on Google Cloud Platform products and collaborated with multiple product and engineering teams.

Software Engineering Lead

JPMORGAN CHASE & CO.

Dec 2016 – May 2021

Hyderabad, Telangana, India

Software Engineering Lead / Architect for an org-wide coding-less Automation Suite for Production Management. Led a team of developers working with Ansible, Netflix Conductor, Spring Boot, Docker / Kubernetes, and RESTful APIs.

Senior Consultant

VIRTUSA POLARIS

Apr 2016 – Dec 2016

Hyderabad, Telangana, India

Senior System Consultant and project lead for 10 associates, organized into Application Support, DBA Support, and Server Support teams providing complete end-to-end support for all Alere Apollo Platform applications.

Senior System Analyst – Project Lead

VALUELABS

Jan 2014 – Apr 2016

Malaysia

Project lead for 7 projects, including 5 support projects with regular monitoring and change-request handling.

- ▶ WebSphere DataPower Migration Lead — led the migration of services from Xi50 DataPower to the latest Xi52 DataPower appliances.
- ▶ Architect / designer for the PDPA project — developed the entire application single-handedly, applying Java object-oriented design principles.

Analyst – Systems Development

VERIZON DATA SERVICES INDIA

Jan 2012 – Jan 2014

Hyderabad, Telangana, India

Analyst – Systems Development, Secondary Release Lead, and Module Lead.

- ▶ Gathered requirements from onsite coordinators and delegated to / worked with the offshore team for successful deliveries.
- ▶ Participated effectively in design and implementation discussions; provided production and testing support whenever required.
- ▶ Testing, debugging, defect fixing, and performance tuning of the codebase; worked closely with CVS & SVN to avoid merge conflicts and keep the history clean.

Systems Engineer

TATA CONSULTANCY SERVICES

Feb 2009 – Dec 2011

Chennai, Tamil Nadu, India

Developer and Backup Project Lead for a team of 7 at offshore.

OPEN SOURCE & INDEPENDENT WORK

Concierge Agent · Agentic AI Platform

Registry-driven multi-agent platform — runtime MCP tool ingestion, tri-layer tool/skill/sub-agent registries, DAG sub-agent workflows, human-in-the-loop approvals, hybrid BM25+embedding retrieval (RRF), OpenTelemetry + LangSmith run tracing (Python · FastAPI · LangGraph · Postgres · React).

J.A.R.V.I.S · Local-First Agent OS

Personal AI operating system — gated policy → approval → execution → verification agent runtime, MCP host with trust-gated tools, provider-agnostic model gateway, AES-256-GCM encrypted memory with pgvector semantic recall, offline voice pipeline, whole-stack acceptance harness (TypeScript · Python · Next.js · Postgres).

MAOS · Multi-Agent Orchestration

Config-driven multi-agent orchestration: LangGraph supervisor, MCP tool plane, A2A agent discovery, OpenAI-compatible API.

Javis-LG · Supervisor–Worker Orchestration

Six-layer LangGraph supervisor–worker system: DB-persisted "living plans," reflection quality loop, dynamic worker registry with per-worker MCP servers, OpenAI-compatible API.

Trilayer Generic Search · Hybrid Retrieval / GraphRAG

Plugin-based retrieval framework fusing FAISS vector, BM25 keyword, and Neo4j graph search via reciprocal rank fusion — LLM intent parsing, grounded citation-constrained synthesis, LLM-as-judge evaluation.

Transcript Intelligence · Applied AI

LLM-hybrid analytics over 100 B2B meeting transcripts with a five-tool MCP server; quantified evaluation — 97% categorization accuracy, r=0.94 sentiment validation, churn-ranking sensitivity analysis.

Whisper Flow Local · Local Voice AI

Fully local, privacy-first dictation — hotkey → local Whisper-family STT → Ollama LLM cleanup → text injection, per-app tone profiles, correction learning, live streaming overlay; MIT-licensed, no cloud.

MLX Fine-Tuning Pipelines · Local ML

LoRA/QLoRA fine-tuning of open-weight models (QwQ-32B, Qwen) on Apple Silicon for enterprise Java codebases — private, zero cloud cost.

MY-Manus · [Agentic AI](#)

Open-source Manus-style AI agent platform on the CodeAct architecture — Docker-sandboxed Python execution, MCP tool discovery, full session replay.

TestRail MCP Server · [AI Tooling](#)

Spring Boot + Spring AI MCP server exposing 101 TestRail operations through 4 Lucene-backed meta-tools with local-only credentials.

Conductor Boot & Starter · [Maven Central](#)

Published artifacts: Spring Boot wrapper of Netflix Conductor Server and a Conductor OSS starter.

Published ML Models · [Hugging Face](#)

StarCoder2-3B (base & instruction-tuned), Flowable documentation models (Llama / Qwen), Netflix Conductor documentation model.

Publications · [Medium](#) / [LinkedIn](#)

"Netflix Conductor — Spring Boot Wrapper" · "Bridging Realms: Humans and AI" · "Scientific Proof of God. Possible?"

CERTIFICATIONS

- ◆ AI Agentic Framework Using Spring AI — Vanderbilt University (In Progress)
 - ◆ Team Software Development with Generative AI — DeepLearning.AI, 2025
 - ◆ Generative AI for Software Development — DeepLearning.AI, 2025
 - ◆ Jenkins, From Zero To Hero: DevOps Jenkins Master — Udemy, 2020
 - ◆ The Complete Agile Scrum Fundamentals Course — Udemy, 2020
- ◆ Spring Boot Microservices and Spring Cloud — Udemy, 2020
 - ◆ Mastering Ansible — Udemy, 2020
 - ◆ IBM WebSphere Certified Solutions Developer: Message Broker — IBM, 2013
 - ◆ IBM WebSphere Certified SOA Associate — IBM, 2013
 - ◆ IBM WebSphere Certified Solutions Developer: Web Services Development — IBM, 2012
 - ◆ IBM WebSphere Certified Solutions Developer: WebSphere Integration Developer — IBM, 2012

EDUCATION

Bachelor of Engineering in Computer Science

Anna University · 2004 – 2008

LANGUAGES

English — Professional	Hindi — Professional	Telugu — Native
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